

Analytics & Charts Specification

Student OS Documentation — File 11

Dashboard Analytics

Attendance: overall percentage + subject bar chart.

Tasks: completed/pending/overdue counts + donut chart.

Study: daily/weekly study hours + line chart.

Exams: preparation progress + progress bars/bar chart.

Academic performance: historical trend when valid marks data exists.

Chart Rules

Charts must use real application data only.

Show empty states when insufficient data exists.

Provide legends/tooltips and accessible text summaries.

Use responsive containers.

Do not overload one chart with unrelated metrics.

Keep units and date ranges explicit.

Metrics

Attendance % = $\text{attended} / \text{total} \times 100$.

Task completion rate = $\text{completed} / \text{total} \times 100$ when $\text{total} > 0$.

Study hours = sum of valid study-session durations.

Exam preparation = configured progress value or clearly defined computed metric.

Performance

Aggregate analytics server-side when datasets become large. Cache expensive calculations where justified.